Quantum Two-Sample Test for Investment Strategies

David Garvin[†] Oleksiy Kondratyev[‡] Alexander Lipton[§]
Marco Paini[¶]

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Abstract

We demonstrate the benefits of using a quantum algorithm rather than its classical counterpart on one of the most fundamental problems of quantitative finance – classification of probability distributions. This problem has many direct applications to practical financial use cases including time series analysis, detection of structural breaks, and monitoring of alpha decay. We present an efficient quantum two-sample test analogous to the classical maximum mean discrepancy test. Experimental results are obtained on Rigetti's Ankaa-2 quantum computer, applied to a specific instance of the probability distribution classification problem. A comparison with the classical maximum mean discrepancy benchmark is provided. The quantum algorithm performs better in terms of discriminatory power. Additionally, the quantum algorithm scales *linearly* with the number of data samples versus a *quadratic* scaling for the classical benchmark.

1 Introduction

Quantum computing capabilities have demonstrated significant progress in recent times. Achievements such as quantum supremacy (Arute *et al* 2019) showcase instances where quantum computers have tackled computations surpassing the capabilities of classical devices. However, it is crucial to note that these accomplishments do not involve problems of immediate practical utility and there are still challenges that current quantum computers need to overcome. Advancements in addressing

[†]Rigetti Computing, Email: dgarvin@rigetti.com
[‡]Imperial College London, Email: a.kondratyev@imperial.ac.uk
[§]Abu Dhabi Investment Authority (ADIA), Email: alexander.lipton@adia.ae
[¶]Rigetti Computing, Email: mpaini@rigetti.com

imperfect control of quantum systems and mitigating noise from environmental interactions are central to the development of quantum computing. Despite the limitations, researchers are actively exploring techniques and applications that could leverage existing quantum computing devices for practical advantages. This pursuit aims to harness the unique capabilities of quantum computers to address real-world problems efficiently and effectively.

In this paper we demonstrate the benefits of using a quantum algorithm rather than its classical counterpart, on one of the most fundamental problems of quantitative finance – classification of probability distributions. The problem can be formulated as follows. Let us assume that we have two sets of samples (either ordered or not), in the most general case of unequal size, drawn from unknown multivariate probability distributions. Can we say with the desired degree of confidence whether these samples were drawn from the same probability distribution or not? This problem has many direct applications to practical use cases, especially on the buy-side, such as time series analysis, detection of structural breaks, and monitoring of alpha decay, to name just a few. The proposed approach draws upon concepts from both classical and quantum machine learning, which are elaborated upon in the subsequent sections.

The main contributions of this paper are as follows:

- An efficient quantum two-sample test analogous to the classical maximum mean discrepancy test (Gretton *et al* 2012).
- The experimental demonstration of the quantum two-sample test method on Rigetti's Ankaa-2 quantum computer, applied to a specific instance of the probability distribution classification problem. A comparison with the classical maximum mean discrepancy benchmark is provided.

The quantum algorithm scales *linearly* with the number of data samples versus a *quadratic* scaling for the classical benchmark.

Section 2 contains background information on relevant basic concepts of quantum computing. Section 3 introduces quantum machine learning and the use of parameterised quantum circuits. Section 4 presents the classical two-sample test used as a benchmark. Section 5 provides details on the new quantum two-sample test. Section 6 contains experimental results. Conclusions are given in Section 7. Appendices contain information on quantum gates and the calculation of expectation values, quantum feature maps, and estimation of the Frobenius distance using a quantum computer.

2 Preliminaries

A computation can be defined as a transformation of one memory state into another, or as a function that transforms information. In quantum computing, the fundamental unit of information is a *qubit* – a quantum mechanical isolated two-level system that expands upon the concept of a classical *bit* (binary digit).

The postulates of quantum mechanics assert that the state of such a system can be represented mathematically by a unit vector $|\psi\rangle$, referred to as *state vector*, in a two-dimensional complex vector space:

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad \alpha, \beta \in \mathbb{C}. \tag{1}$$

In general, the state $|\psi\rangle$ is a linear combination (*superposition*) of the orthogonal *standard computational basis vectors*:

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad |\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,. \tag{2}$$

Since the state vector is a unit vector, the coefficients α and β must satisfy

$$|\alpha|^2 + |\beta|^2 = 1. \tag{3}$$

The coefficients α and β are *probability amplitudes*. Even though a qubit can exist in a superposition of computational basis states, once *measured* in the computational basis, its state *collapses* to one of the computational basis states; $|\alpha|^2$ and $|\beta|^2$ give us the probability of finding the qubit, respectively, in states $|0\rangle$ and $|1\rangle$ after measurement. The symbols $|\psi\rangle\,, |0\rangle\,, |1\rangle$ come from traditional physics notation, where the "ket", denoted as $|x\rangle$, represents the column vector with elements $(x_1, \ldots, x_N)^\top$, and the "bra", denoted as $\langle x|$, represents the row vector with complex conjugate elements $(x_1^*, \ldots, x_N^*)$. A system of n qubits has an associated vector space given by the tensor product of the vector spaces of the individual qubits, thus with dimension 2^n, and its state is represented by a unit vector of such space.

If a multi-qubit state can be expressed as a tensor product of individual qubit states, e.g., $|\psi_{12}\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$, then the qubits are said to be separable (measured quantities on the individual qubits are uncorrelated). Otherwise, the qubits are *entangled* (measured quantities on the individual qubits are correlated). A system of n separable qubits can be described by specifying $2n$ probability amplitudes; if the qubits are entangled, there can be 2^n independent probability amplitudes (up to the normalisation condition – the state needs to be of length 1). Therefore, a generic multi-qubit state, even only with a few tens of qubits, can encode in principle a large amount of information.

Operations that transform qubit states are called *quantum logic gates*. An n-qubit gate is a linear unitary operator in the space of the n qubits which can be represented as a $2^n \times 2^n$ unitary matrix. A unitary matrix preserves the norm and maps a unit vector into another unit vector. A sequence of n-qubit gate followed by measurement completes a cycle in a quantum computation.

In classical computing there are only two possible 1-bit logic gates: an identity gate that leaves the bit state unchanged and a NOT gate that flips the bit state. In contrast, there are infinitely many possible 1-qubit quantum logic gates (2×2 unitary matrices). Some of them, such as Pauli gates, play special roles. Appendix A provides detailed coverage of notable examples of quantum gates.

Example 1: We can easily verify by direct calculations that the Pauli X gate implements a quantum NOT that flips the qubit state:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \tag{4}$$

while the Pauli Z gate implements a PHASE gate that flips the relative phase:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = - \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{5}$$

Finally, we need to introduce the *density matrix*. Let $\langle \psi_i | \psi_j \rangle$ and $|\psi_i\rangle \langle \psi_j|$ denote, respectively, the inner and the outer products of two state vectors. A density operator ρ is a linear positive semidefinite Hermitian operator with unit trace and can in general be expressed as

$$\rho = \sum_{k=1}^{K} p_k |\psi_k\rangle \langle \psi_k|, \quad \sum_{k=1}^{K} p_k = 1, \quad p_k \geq 0, \text{ for all } k = 1, \ldots, K. \tag{6}$$

When the state of a system is $|\psi_1\rangle$, then the density operator is simply $|\psi_1\rangle \langle \psi_1|$ and the state is *pure*. If the system is in $|\psi_1\rangle$ with probability p_1, $|\psi_2\rangle$ with probability p_2, etc., then the state is *mixed* and can only be represented through the density operator above. In matrix form, the linear density operator ρ corresponds to a density matrix such that

$$\rho = \rho^\dagger, \quad \mathrm{Tr}(\rho) \equiv \sum_{k=1}^{K} \rho_{kk} = 1, \quad \rho_{kk} \geq 0, \text{ for all } k = 1, \ldots, K, \tag{7}$$

where the symbol ρ has been used for both the density operator and matrix.

Example 2: An example of a pure state is one that can be built with the Hadamard state $|-\rangle$:

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \tag{8}$$

with corresponding density matrix

$$\rho = |-\rangle \langle -| = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \tag{9}$$

Example 3: An example of a mixed state is a *statistical ensemble* of states $|0\rangle$ and $|1\rangle$. If a physical system is prepared to be either in state $|0\rangle$ or state $|1\rangle$ with equal probability, it can be described by the *maximally mixed* state

$$\rho = \frac{1}{2} |0\rangle \langle 0| + \frac{1}{2} |1\rangle \langle 1| = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \tag{10}$$

Note that this is different from the density matrix of the pure state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \tag{11}$$

which is an equal superposition of states $|0\rangle$ and $|1\rangle$, and which reads

$$\begin{aligned}\rho_\psi = |\psi\rangle\langle\psi| &= \frac{1}{2}(|0\rangle + |1\rangle)(\langle 0| + \langle 1|) \\ &= \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 1|) = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.\end{aligned} \tag{12}$$

3 Quantum Machine Learning

The new computational paradigm of quantum computing prompted extensive research in the emerging field of quantum machine learning (QML). Similar to their classical counterparts, QML models can be implemented with a supervised, unsupervised and reinforcement learning approach, and can be utilised for both discriminative and generative tasks (Biamonte *et al* 2017). A common encoding approach in QML is to use parameterised quantum circuits (PQC), given the general possibility – different in the details for each quantum computing platform – to execute parameterised quantum gates. When the parameters are utilised as the variables of a loss objective function, the PQC becomes analogous to a classical neural network.

Figure 1 shows a schematic representation of a PQC composed of single-qubit and 2-qubit quantum gates. The final quantum state $|\psi_f\rangle$ is obtained after the application of the n-qubit quantum circuit – a sequence of quantum logic gates (linear unitary operators $\{U_i\}$) fully defined by the parameters $\theta_1, \ldots, \theta_m$ – to the initial quantum state $|\psi_0\rangle$:

$$|\psi_f\rangle = U_m(\theta_m) \ldots U_2(\theta_2) U_1(\theta_1) |\psi_0\rangle. \tag{13}$$

The final state can be measured and information on $|\psi_f\rangle$ can be extracted.

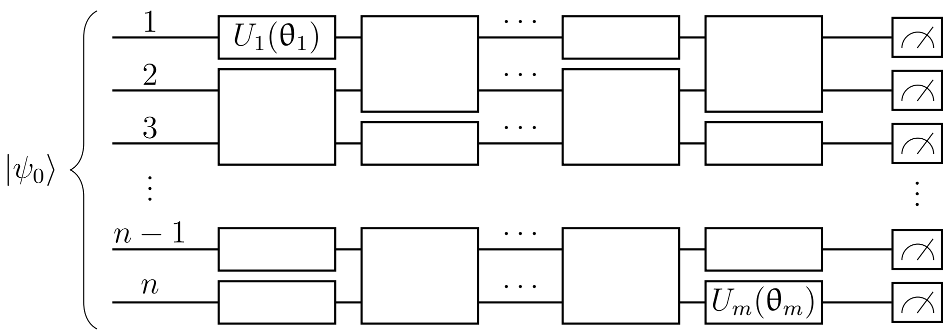

Figure 1: Schematic representation of a parameterised quantum circuit on n qubits.

A PQC can encode classical data samples into a space of exponentially larger dimension. The map from a set of data points to a set of quantum states through a PQC is called a *quantum feature map*. There are multiple possible schemes to map the vector of classical features $\mathbf{x}$ into the PQC parameters θ, a popular approach being the *angle encoding* method (see, e.g., (Jacquier and Kondratyev 2022)). PQCs have received significant interest because of their generality and expressive power (Benedetti *et al* 2019). For example, (Du *et al* 2018) show that a specific class of generative quantum machine learning models has strictly more expressive power than the class of analogous classical models, when only a polynomial number of parameters are allowed.

4 Classical Two-Sample Test

For multivariate distributions, a widely used measure of similarity is the maximum mean discrepancy (MMD), which can be used in a kernel based two-sample statistical test to evaluate whether two distributions are the same. MMD can also be used as a cost function in various generative machine learning models (Gretton *et al* 2012).

The MMD approach to two-sample testing is defined by the idea of representing distances between distributions as distances between mean embeddings of features. If we have distributions P and Q over a set $\mathcal{F}$, and a feature map $\varphi : \mathcal{F} \to \mathcal{H}$, with $\mathcal{H}$ a reproducing kernel Hilbert space, the MMD between P and Q can be defined as

$$\mathrm{MMD}(P, Q) = \|\mathbb{E}_{\mathbf{x}\sim P}[\varphi(\mathbf{x})] - \mathbb{E}_{\mathbf{y}\sim Q}[\varphi(\mathbf{y})]\|_{\mathcal{H}}. \tag{14}$$

Example 4: $\mathcal{F} = \mathcal{H} = \mathbb{R}^d$ and $\varphi(\mathbf{x}) = \mathbf{x}$. In this example, the MMD is just the distance between the means of the two distributions:

$$\mathrm{MMD}(P, Q) = \|\mathbb{E}_{\mathbf{x}\sim P}[\mathbf{x}] - \mathbb{E}_{\mathbf{y}\sim Q}[\mathbf{y}]\|_{\mathbb{R}^d} = \|\mu_P - \mu_Q\|_{\mathbb{R}^d}. \tag{15}$$

Example 5: The linear transformation of Example 4 could be generalised with $\mathcal{F} = \mathbb{R}^d$, $\mathcal{H} = \mathbb{R}^p$ and $\varphi(\mathbf{x}) = \mathbf{A}\mathbf{x}$, where $\mathbf{A}$ is a $p \times d$ matrix:

$$\begin{aligned}\mathrm{MMD}(P, Q) &= \|\mathbb{E}_{\mathbf{x}\sim P}[\mathbf{A}\mathbf{x}] - \mathbb{E}_{\mathbf{y}\sim Q}[\mathbf{A}\mathbf{y}]\|_{\mathbb{R}^p} \\ &= \|\mathbf{A}\mathbb{E}_{\mathbf{x}\sim P}[\mathbf{x}] - \mathbf{A}\mathbb{E}_{\mathbf{y}\sim Q}[\mathbf{y}]\|_{\mathbb{R}^p} = \|\mathbf{A}(\mu_P - \mu_Q)\|_{\mathbb{R}^p}.\end{aligned} \tag{16}$$

Example 6: $\mathcal{F} = \mathbb{R}^1$ and $\mathcal{H} = \mathbb{R}^2$ with $\varphi(x) = (x, x^2)$:

$$\mathrm{MMD}(P, Q) = \sqrt{(\mathbb{E}_{x\sim P}[x] - \mathbb{E}_{y\sim Q}[y])^2 + (\mathbb{E}_{x\sim P}[x^2] - \mathbb{E}_{y\sim Q}[y^2])^2}. \tag{17}$$

This MMD represents a *stronger* distance than simply the difference in the means, since it would discriminate not only between distributions with different means but also with different variances.

Given MMD only depends on linear products of φ, we can compute MMD using the *kernel trick* (Muandet *et al* 2017)

$$\langle \varphi(\mathbf{x}), \varphi(\mathbf{y}) \rangle_{\mathcal{H}} = k(\mathbf{x}, \mathbf{y}), \tag{18}$$

with positive semidefinite *kernel function* $k(\cdot,\cdot)$:

$$\begin{aligned}\mathrm{MMD}^2(P,Q) &= \|\mathbb{E}_{\mathbf{x}\sim P}[\varphi(\mathbf{x})] - \mathbb{E}_{\mathbf{y}\sim Q}[\varphi(\mathbf{y})]\|^2_{\mathcal{H}} \\ &= \mathbb{E}_{\mathbf{x},\mathbf{x}'\sim P}k(\mathbf{x},\mathbf{x}') - 2\mathbb{E}_{\mathbf{x}\sim P,\mathbf{y}\sim Q}k(\mathbf{x},\mathbf{y}) + \mathbb{E}_{\mathbf{y},\mathbf{y}'\sim Q}k(\mathbf{y},\mathbf{y}').\end{aligned} \tag{19}$$

MMD can be estimated directly from a finite set of samples. For two datasets $\mathcal{X}$ and $\mathcal{Y}$ with cardinality N and M respectively, MMD can be estimated using (Gretton *et al* 2012)

$$\begin{aligned}\mathrm{MMD}^2 = \frac{1}{N^2}\sum_{i,j=1}^{N} k(\mathbf{x}_i,\mathbf{x}_j) - \frac{2}{NM}\sum_{i=1}^{N}\sum_{j=1}^{M} k(\mathbf{x}_i,\mathbf{y}_j) \\ + \frac{1}{M^2}\sum_{i,j=1}^{M} k(\mathbf{y}_i,\mathbf{y}_j).\end{aligned} \tag{20}$$

When the map $P \to \mathbb{E}_{\mathbf{x}\sim P}[\varphi(\mathbf{x})]$ is injective, the associated kernel is said to be characteristic. A well-known example of a characteristic kernel is the Gaussian kernel

$$k_\xi(\mathbf{x}_i,\mathbf{x}_j) = \exp\left\{-\frac{\|\mathbf{x}_i-\mathbf{x}_j\|^2}{2\xi^2}\right\} = \exp\left\{-\frac{\mathbf{x}_i^\top\mathbf{x}_i - 2\mathbf{x}_i^\top\mathbf{x}_j + \mathbf{x}_j^\top\mathbf{x}_j}{2\xi^2}\right\}, \tag{21}$$

where ξ is a bandwidth parameter. For a characteristic kernel, $\mathrm{MMD}(P,Q)$ is equal to zero if and only if $P = Q$. In other terms, the MMD test with a characteristic kernel, such as the Gaussian kernel, is exact up to statistical errors due to finite sampling. This motivates our choice of a Gaussian kernel as the classical benchmark for the proposed quantum algorithm.

5 Quantum Two-Sample Test

Having introduced the fundamental tools that we need for the quantum algorithm, let us go back to the problem we aim to solve.

Let A and B be two datasets, sampled, respectively, from the probability distributions $P(\mathbf{x})$ and $Q(\mathbf{x})$, $\mathbf{x} \in \mathcal{D} \subseteq \mathbb{R}^m$. The cardinality of A and B will be indicated as $[A]$ and $[B]$. The samples $\mathbf{a} \in A$ and $\mathbf{b} \in B$ are of the form $\mathbf{a}_i := (a_i(1),\dots,a_i(m))^\top$ and $\mathbf{b}_j := (b_j(1),\dots,b_j(m))^\top$. We define the feature map φ as a function that maps the samples $\mathbf{a}_i$ and $\mathbf{b}_j$ into vectors $\varphi(\mathbf{a}_i)$ and $\varphi(\mathbf{b}_j)$ of a Hilbert space $\mathcal{H}_{\mathcal{O}}$. In the proposed quantum algorithm, the feature map φ is realised via a unitary transformation $U(\mathbf{x})$ applied to the initial state $|\mathbf{0}\rangle \equiv |0\rangle^{\otimes n}$,

$$\varphi(\mathbf{x}) = U(\mathbf{x})\,|\mathbf{0}\rangle\,\langle\mathbf{0}|\,U(\mathbf{x})^\dagger \equiv |\mathbf{x}\rangle\,\langle\mathbf{x}|\,. \tag{22}$$

Note that if the transformation φ is specified on n qubits, then the dimension of the Hilbert space $\mathcal{H}_{\mathcal{O}}$ of linear operators is $4^n = 2^n \times 2^n$.

We construct two density matrices defined as sample means of $P(\mathbf{x})$ and $Q(\mathbf{x})$:

$$p = \sum_{i=1}^{[A]} \frac{1}{[A]} |\mathbf{a}_i\rangle \langle \mathbf{a}_i| \,, \quad q = \sum_{j=1}^{[B]} \frac{1}{[B]} |\mathbf{b}_j\rangle \langle \mathbf{b}_j| \,. \tag{23}$$

The density matrices p and q and the projectors $|\mathbf{x}\rangle \langle \mathbf{x}|$ that form them are all vectors in the space $\mathcal{H}_\mathcal{O}$. We introduce a distance d in $\mathcal{H}_\mathcal{O}$ and use it as measure of closeness of p and q. As for the classical MMD, this is a proxy for whether A and B are sampled from the same probability distribution.

In the limit case of infinitely many samples, for an injective $U(\mathbf{x})$, it is easy to see how $d(p,q) = 0$ if and only if $P(\mathbf{x}) = Q(\mathbf{x})$. In fact,

$$\begin{aligned} p - q &= \int_\mathcal{D} d\mathbf{x} P(\mathbf{x}) |\mathbf{x}\rangle \langle \mathbf{x}| - \int_\mathcal{D} d\mathbf{x} Q(\mathbf{x}) |\mathbf{x}\rangle \langle \mathbf{x}| \\ &= \int_\mathcal{D} d\mathbf{x} (P(\mathbf{x}) - Q(\mathbf{x})) |\mathbf{x}\rangle \langle \mathbf{x}| \end{aligned} \tag{24}$$

is equal to 0 if and only if $P(\mathbf{x}) - Q(\mathbf{x}) = 0, \forall \mathbf{x}$, or, given the assumed injectivity of U, if and only if $P(\mathbf{x}) = Q(\mathbf{x}), \forall \mathbf{x}$. To conclude that $d(p,q) = 0$ if and only if $P(\mathbf{x}) = Q(\mathbf{x}), \forall \mathbf{x}$, we only need to remember that for any distance d, the conditions $d(p,q) = 0$ and $p = q$ are equivalent.

In the case of finite A and B, a distance $d > 0$ would in general be measured even if A and B contained elements sampled from $P(\mathbf{x})$ and $Q(\mathbf{x})$ with $P = Q$. This discrepancy from $d = 0$ grows with increasing variance of the distribution and diminishes with increasing cardinality of the datasets.

The proposed quantum two-sample test is an MMD test. In fact, the distance between density matrices is the maximum mean discrepancy with feature map defined by a quantum transformation. Predicting the a priori effectiveness of an MMD test for a specific choice of a feature map is challenging (Muandet *et al* 2017). In the upcoming section, we will directly compare results obtained using a quantum feature map against those derived from classical feature maps, focusing on a specific problem of interest.

The distance that we will consider in the experiments that will follow is the Frobenius distance

$$d_F(p,q) = \frac{1}{2}\|p - q\| = \frac{1}{2}\sqrt{\mathrm{Tr}[(p-q)^2]}. \tag{25}$$

To summarise, our task is two-fold:

1. We need to construct the quantum feature map that will encode the classical datasets into the density matrices.
2. We need to calculate the Frobenius distance between the density matrices.

The proposed approach is as follows:

1. Classical samples are mapped into quantum states with the help of a PQC,

with layers of 1-qubit gates used to encode the data alternating with layers of 2-qubit gates – see Appendix B.

2. The Frobenius distance between two density matrices is calculated using "approximate quantum states" (Paini *et al* 2021) – see Appendix C.

6 Experimental Results

We conducted experiments on Ankaa-2 – the 4th generation superconducting quantum processor released by Rigetti Computing in 2023.

The datasets we use to illustrate the performances of the classical and the quantum models consist of three simulated time series A, B and C. The time series represents the asset price process, S_t, with variable degree of serial dependence controlled by the autocorrelation parameter ϱ:

$$S_t = S_{t-1} \exp\left\{ \left(\mu - \frac{1}{2}\sigma^2\right) \Delta t + \sigma\sqrt{\Delta t}\left(\varrho z_{t-1} + \sqrt{1-\varrho^2} z_t\right) \right\}, \qquad z_{t-1}, z_t \sim N(0,1). \tag{26}$$

Here, μ is the annualised drift (set equal to 0.1), σ is the annualised volatility (set equal to 0.15), Δt is the year fraction (set equal to $1/250 = 1$ business day). The autocorrelation was set at $\varrho = 0$ for asset A, $\varrho = 0.2$ for asset B, and $\varrho = 0.4$ for asset C. The simulation is run for 5 years (1,250 observations). Figure 2 displays sample generated datasets. The task is to quantify the differences between the time series A, B and C and decide whether the null hypothesis of the time series being drawn from the same probability distribution can be rejected at the given confidence level.

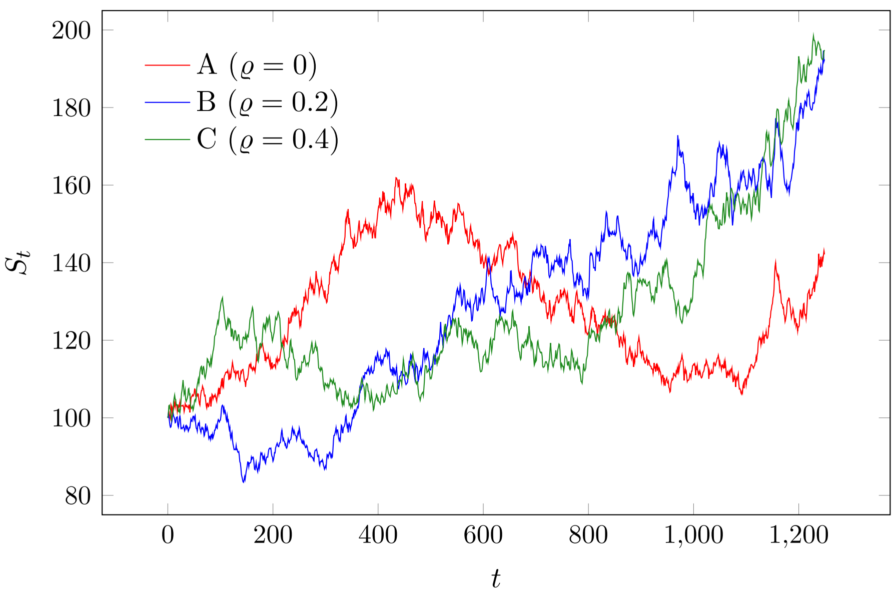

Figure 2: Sample generated dataset.

Note that the univariate distribution of daily log-returns is the same for all assets. This is because the stochastic driver of daily log-returns,

$$\varrho z_{t-1} + \sqrt{1-\varrho^2} z_t, \tag{27}$$

is normally distributed with mean of 0 and variance 1 for all values of ρ. We can also verify it numerically by running the univariate Kolmogorov-Smirnov two-sample test on the time series of log-returns for assets A, B and C. The Kolmogorov-Smirnov test would fail to discriminate between the time series A, B and C since it ignores any possible serial dependence within the time series. It assumes that the samples (in our case daily log-returns) are independent and identically distributed (i.i.d.) random variables. However, knowledge of the general characteristics of financial time series suggests that we should pay attention to several important aspects of the specific problem.

Firstly, financial asset returns are rarely, if ever, iid. Therefore, we must construct several features that can capture known effects such as a serial dependence of returns (an up move is more likely to be followed by another up move) and a serial dependence of the absolute values of returns (a large amplitude move is more likely to be followed by another large amplitude move). To address these points, we construct the day t sample, $\mathbf{F}_t$, as a vector of the following four features:

$$F_t(1) = \ln\left(\frac{S_t}{S_{t-1}}\right), \quad F_t(2) = \ln\left(\frac{S_{t-1}}{S_{t-2}}\right), \tag{28}$$

$$F_t(3) = \left|\ln\left(\frac{S_t}{S_{t-1}}\right) + \ln\left(\frac{S_{t-1}}{S_{t-2}}\right)\right|, \quad F_t(4) = \left|\ln\left(\frac{S_t}{S_{t-1}}\right) - \ln\left(\frac{S_{t-1}}{S_{t-2}}\right)\right|.$$

The choice of features is motivated by the following considerations. If we set $\rho = 0$, we end up with the process without any serial dependence:

$$S_t = S_{t-1} \exp\left\{\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} z_t\right\}. \tag{29}$$

In this case the log-return $\ln(S_t/S_{t-1})$ is only driven by z_t and is independent of the previous log-return $\ln(S_{t-1}/S_{t-2})$ which is only driven by z_{t-1}.

However, for any $\rho > 0$, the log-returns $\ln(S_t/S_{t-1})$ and $\ln(S_{t-1}/S_{t-2})$ are no longer independent. This is because they share the same stochastic driver. Log-return $\ln(S_t/S_{t-1})$ is driven by z_t and z_{t-1} while log-return $\ln(S_{t-1}/S_{t-2})$ is driven by z_{t-1} and z_{t-2}. The shared stochastic driver is z_{t-1}, which enters the expression for S_t with coefficient ρ.

Secondly, since the day t sample contains day t and day $t-1$ log-returns, we can only use 625 non-overlapping samples from the time series of 1,250 daily observations.

Finally, in order to produce robust estimates of the statistical errors, we need to run multiple simulations of the time series.

Figure 3 shows the MMDs obtained with a classical Gaussian kernel and with a quantum feature map, executed on a quantum computer simulator and on Rigetti Ankaa-2, for all index pairs. The quantum circuit uses four qubits and is specified in Appendix B. The algorithm and Frobenius distance details are in Appendix C.

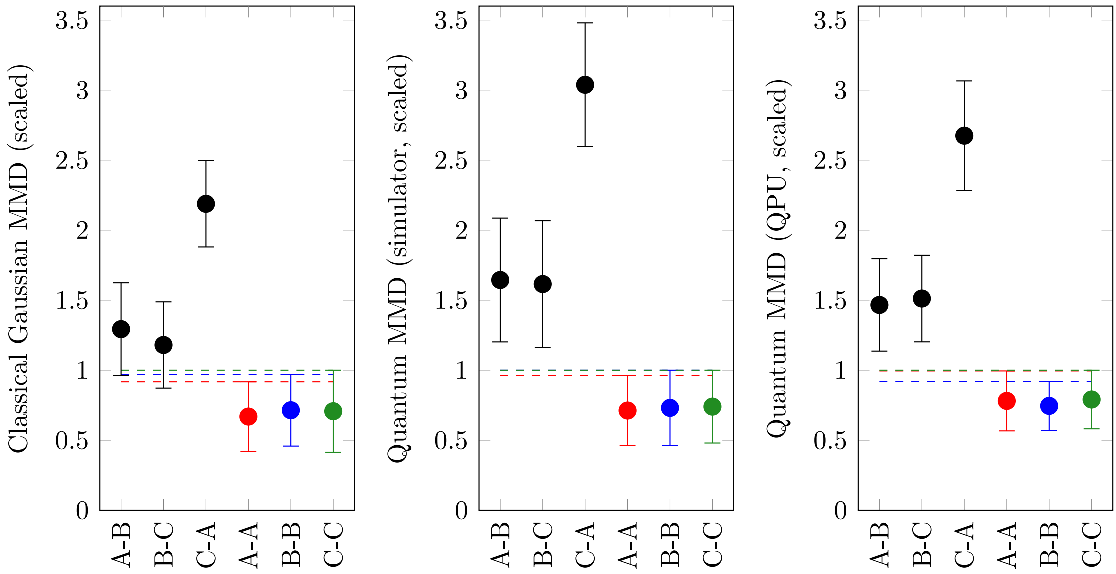

Figure 3: Comparison of classical (left) and quantum distances for 100 simulated datasets A ($\varrho = 0$), B ($\varrho = 0.2$) and C ($\varrho = 0.4$). The quantum results are obtained using a parameterised quantum circuit (Appendix B) on a simulator (center) or ring of qubits $\{63, 64, 71, 70\}$ on the Rigetti Ankaa-2 quantum processor (QPU) (right).

The classical and quantum MMDs were calculated for 100 simulated scenarios for assets A, B and C. The black dots show the mean distances between two different asset datasets for all asset pairs and the black error bars display ± 1 standard deviation around the mean values across all simulated scenarios. On their own, these quantities can say only how close (or distant) the two asset datasets are relative to the distances between other asset pairs. To deduce whether the two given asset time series were drawn from the same probability distribution, we need to estimate the natural variance of the samples drawn from this probability distribution. This is what is visually represented by the coloured dots and error bars. The coloured dots show the mean distances between the various realisations of the same asset dataset across all simulated scenarios and the coloured error bars show the ± 1 standard deviation intervals around the mean values. The ratios of mean distances are presented in Table 1.

The error bars reflect the amount of noise present in the estimates of classical and quantum MMD. The accuracy of the estimates depends on the number of samples in the dataset (the length of the error bars) and the number of simulations (uncertainty in the length of the error bars). A general rule suggests that the noise should decrease as $\mathcal{O}(1/\sqrt{N})$, where, depending on the context, N is the number of samples in the dataset or the number of simulations performed.

Ratio	Classical	Simulator	QPU
mean(A-B) / mean(A-A)	1.9 ± 0.1	2.3 ± 0.1	1.9 ± 0.1
mean(A-B) / mean(B-B)	1.8 ± 0.1	2.2 ± 0.1	2.0 ± 0.1
mean(B-C) / mean(B-B)	1.7 ± 0.1	2.2 ± 0.1	2.0 ± 0.1
mean(B-C) / mean(C-C)	1.7 ± 0.1	2.2 ± 0.1	1.9 ± 0.1
mean(C-A) / mean(C-C)	3.1 ± 0.2	4.1 ± 0.2	3.4 ± 0.1
mean(C-A) / mean(A-A)	3.3 ± 0.2	4.3 ± 0.2	3.4 ± 0.1

Table 1: Classical versus quantum distances (with 1 standard deviation sample errors).

In Figure 3, MMD with classical Gaussian kernel allows to clearly discriminate between the time series of assets A ($\varrho = 0$) and C ($\varrho = 0.4$). For the two other asset pairs, A-B and B-C, the 1-standard deviation confidence intervals overlap and it is impossible to say with certainty whether the time series of asset returns were drawn from two different distributions. Note that the bandwidth parameter ξ of the Gaussian kernel used was equal to 1. A parameter search did not point to values of ξ that would lead to relevant greater discriminatory power of the MMD test. In contrast, the quantum MMD results show that their confidence intervals do not overlap and enable clear discrimination of asset pairs A-B and B-C.

In general, the time series cannot be adequately represented by a univariate or even a low-dimensional multivariate distribution. A univariate distribution may provide information about the first, second, third, etc. moments but will always fail to capture patterns presented in the sequence of samples. We need to construct multiple features (i.e., a multivariate distribution) to describe the time series adequately.

Figure 3 illustrates the quantum algorithm performing better in terms of discriminatory power. Clear gaps occur between the black dots' error bars and the horizontal dashed lines of the coloured dots' error bars. There are also noticeably larger ratios of mean distances. The result on the QPU is inferior to the simulated result, as the QPU errors, often referred to as noise, negatively affect the outcome. However, it is important to observe that the detrimental impact of noise is somewhat limited and the discrimination between pairs of distributions is still effective. This is the result of a number of factors. The algorithm is inherently robust to "linear" errors, i.e., errors that only reduce expectation values by a factor – provided that the factor is not excessively small – since the test is based on relative, not absolute, values. The approximate state technique used for the experiment converts measurement errors into approximately linear errors (Arrasmith *et al* 2023) and the limited depth of the circuit accounts for the limited contribution of gate errors to the deviation from the simulated result.

Furthermore, it should be noted that the number of core function calls (Gaussian kernel function) scales *quadratically* with the number of samples for the Gaussian kernel, unlike the quantum case, which requires only a linear number of core function executions. Specifically, the number of quantum circuit runs and the Frobenius distance estimation algorithm (Appendix C) scale *linearly* with the number of samples. Although in practice, due to the statistical nature of quantum computation, the quantum circuit (Appendix B) must be run multiple times for every sample, with 2-qubit gate times of approximately $60 - 70$ nanoseconds (Rigetti 2024), and overall circuit execution times in the order of a few microseconds, an execution time advantage of quantum MMD compared to a classical kernel MMD may begin to emerge for datasets with a number of samples larger than the number of quantum circuit runs per sample. However, for a comprehensive comparison of classical and quantum execution times, it is important to note that there are classical approximations of Gaussian kernels, called random Fourier features (Muandet *et al* 2017), that only require a linear number of function calls. Although random Fourier features are only approximations of Gaussian kernels and may generally exhibit lower discriminatory power, their MMD capabilities for the use case of this paper have not been explicitly measured.

As a final observation on the experiment conducted, we note that the comparative approach employed (comparing the mean of (A-B) to the means of (A-A) and (B-B)) was facilitated by our ability to run multiple simulations of the time series. If, instead, we were given two time series without the capability to regenerate points from the same distributions, the statistical uncertainties arising from finite sampling could have been addressed using a generalised χ^2-like test approach based on the confidence intervals of the square norm of the quantum features. For non-stationary processes, finite distances could be interpreted as changes in the process regimes.

7 Conclusions and Outlook

In this paper we presented a quantum computing algorithm – albeit at small qubit count and low circuit depth – outperforming its classical benchmark on a specific real-world problem (classification of multivariate probability distributions) through higher discriminatory power and improved scaling in the number of samples.

The proposed algorithm achieves quadratic speedup in the number of key function calls since it scales linearly with the number of samples in the dataset in contrast with the classical benchmark – MMD with Gaussian kernel – which is quadratic in the number of samples. This may become a material advantage for large datasets as quantum computing technology matures.

In finance, the main applications are likely to be found on the buy-side. Being able to clearly identify two time series as coming from two different distributions would allow us to identify a structural break between in-sample and out-of-sample investment strategy performance (graduation testing), as well as being used for monitoring of alpha decay (investment strategies in live trading).

Another important application of the proposed quantum two-sample test is related to synthetic data generation. The Frobenius distance between two density matrices can be used as a natural discriminator between the original (training) and the generated (synthetic) datasets.

The quantum algorithm can be improved in several ways. In particular, the time dependence inherent in the datasets could be more explicitly exploited in the quantum feature map, utilising a circuit that acts as a dynamic reservoir (Xiong *et al* 2023). Furthermore, the quantum feature map could be combined with signature kernels (Paini *et al* 2023) to then compute distances in signature space. The performances of circuits with greater depth and greater quantum complexity should also be explored. Error-mitigated deeper circuits, coupled with the higher qubit count necessary for problems with a large number of features, will pose challenges to classical simulation and underscore the necessity for execution exclusively on quantum hardware.

One of the possible directions of further research would be to test the power of the proposed method under heteroscedasticity which often represents more real market scenarios, e.g., fat tailed distributions. Empirical results suggest that in such situations classical MMD may fail to capture serial dependence. This opens another exciting avenue of potential quantum MMD applications.

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A Quantum Gates and Expectation Values

The Bloch sphere (Figure 4) visualises the canonical representation of a qubit state as a unit vector in the 2-dimensional complex vector space:

$$|\psi\rangle = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) \\ e^{i\phi}\sin\left(\frac{\theta}{2}\right) \end{bmatrix}. \tag{30}$$

Angles $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi]$ uniquely determine the position of the qubit on the unit sphere. Mathematically, a single qubit gate is any 2×2 unitary matrix that changes the qubit state and which can be visualised as a transition from one point on the surface of the Bloch sphere to another. A unitary matrix preserves the norm and maps a unit vector into another unit vector – this is why the 1-qubit gates are often referred to as *rotations*.

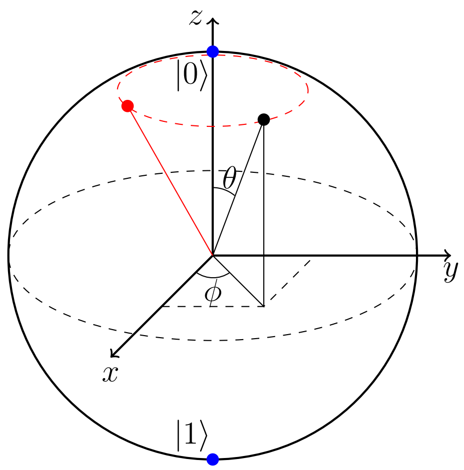

Figure 4: Schematic representation of the Bloch sphere with rotation around the z axis.

In the computational basis (z-basis), the North pole of the Bloch sphere ($\theta = 0$) corresponds to state $|0\rangle$:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \tag{31}$$

and the South pole of the Bloch sphere ($\theta = \pi$) corresponds to state $|1\rangle$:

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \tag{32}$$

and an arbitrary qubit state $|\psi\rangle$ can be represented by a superposition of the basis states $|0\rangle$ and $|1\rangle$

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \quad |\alpha|^2 + |\beta|^2 = 1, \tag{33}$$
$$\alpha = \cos\left(\frac{\theta}{2}\right), \quad \beta = e^{i\phi}\sin\left(\frac{\theta}{2}\right),$$

where α and β are probability amplitudes. The probability of finding the qubit in state $|0\rangle$ ($|1\rangle$) after measurement in the computational basis is given by $|\alpha|^2$ ($|\beta|^2$).

Pauli matrices X, Y and Z are rotations of the qubit state by π radians around, respectively, the x, y and z axes:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \tag{34}$$

Rotations around the x, y and z axes by an arbitrary angle θ are traditionally denoted as $R_x(\theta)$, $R_y(\theta)$ and $R_z(\theta)$. The corresponding unitary matrices are:

$$R_x(\theta) = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right) \\ -i\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}, \quad R_y(\theta) = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}, \tag{35}$$

$$R_z(\theta) = \begin{bmatrix} \mathrm{e}^{-i\theta/2} & 0 \\ 0 & \mathrm{e}^{i\theta/2} \end{bmatrix}.$$

There are more 1-qubit gates that play special roles, for example

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}. \tag{36}$$

The gate I is an identity gate that leaves the qubit state unchanged. The gate H is known as the Hadamard gate and it creates an equal superposition of states $|0\rangle$ and $|1\rangle$ if applied to either state $|0\rangle$ or state $|1\rangle$:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \tag{37}$$

The Hadamard gate also changes the z-basis (standard computational basis) into the x-basis. States $|0\rangle$ and $|1\rangle$ are eigenstates of Pauli Z with eigenvalues $+1$ and -1

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle, \tag{38}$$

and the Hadamard states $|+\rangle$ and $|-\rangle$:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \tag{39}$$

are eigenstates of Pauli X with eigenvalues $+1$ and -1:

$$X|+\rangle = |+\rangle, \quad X|-\rangle = -|-\rangle. \tag{40}$$

Similarly, a combination of H and S gates (with the H gate applied first) changes the z-basis into the y-basis

$$SH|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle), \quad SH|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle). \tag{41}$$

States $|+i\rangle$ and $|-i\rangle$:

$$|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle), \quad |-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle), \tag{42}$$

are eigenstates of Pauli Y with eigenvalues $+1$ and -1:

$$Y\,|+i\rangle = |+i\rangle\,, \quad Y\,|-i\rangle = -\,|-i\rangle\,. \tag{43}$$

Figure 5 shows all the basis states on the Bloch sphere.

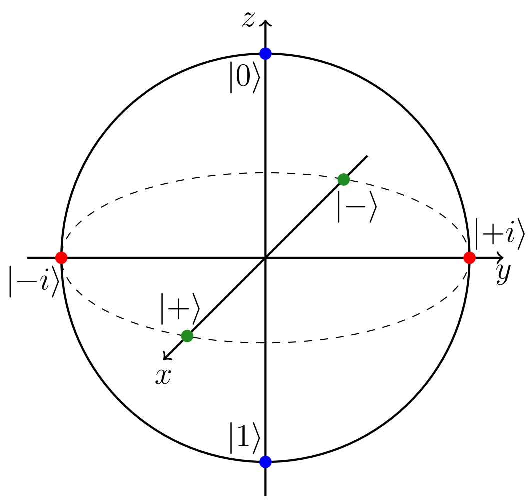

Figure 5: Basis states.

Similar to 1-qubit gates specified by unitary 2×2 matrices, we can construct any number of multi-qubit gates. The n-qubit gates can be represented by $2^n \times 2^n$ unitary matrices. Since multi-qubit gates act on several qubits at the same time, they can be used to entangle them – i.e. make the overall state non-expressible as the product of individual qubit states.

Note that the X gate flips the qubit state. This makes it the quantum NOT gate. A useful 2-qubit gate that is widely used in quantum algorithms is the *controlled NOT* gate, usually denoted as CNOT or CX gate. It consists in applying the X gate to the target qubit if the control qubit is in state $|1\rangle$, otherwise it applies the identity gate I), and is given by the unitary matrix

$$\mathrm{CNOT} \equiv \mathrm{CX} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}. \tag{44}$$

This gate is often represented in the quantum circuit by an XOR (exclusive OR) logical symbol (circled plus) placed on the target qubit quantum register since its truth table (for the target qubit) coincides with the truth table of the XOR logic gate.

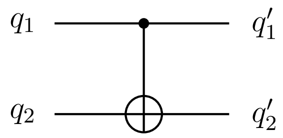

Figure 6: CX (CNOT) gate.

q_1	q_2	q'_1	q'_2
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

Table 2: Truth table for the CX (CNOT) gate.

Another useful controlled gate is the *controlled* Z, usually denoted as CZ. It is the Z gate (phase flip) applied to the target qubit conditional on the control qubit being in state $|1\rangle$ and is given by the unitary matrix:

$$\mathrm{CZ} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}. \tag{45}$$

The SWAP gate swaps the states of two qubits. The $\sqrt{\mathrm{SWAP}}$ gate, in the same way as the CX and CZ gates, is universal, in the sense that any multi-qubit gate can be constructed from only $\sqrt{\mathrm{SWAP}}$ and single qubit gates.

$$\mathrm{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \sqrt{\mathrm{SWAP}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1+i}{2} & \frac{1-i}{2} & 0 \\ 0 & \frac{1-i}{2} & \frac{1+i}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{46}$$

Normally, the choice of the set of physical gates on hardware is dictated by the characteristics of the physical system used to perform the quantum computation. Other common physical gates are:

$$\mathrm{iSWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \sqrt{\mathrm{iSWAP}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{47}$$

CZ and iSWAP are native 2-qubit gates in Rigetti Ankaa systems.

As mentioned, multi-qubit gates can entangle qubits. An n-qubit system can exist in any superposition of the 2^n basis states

$$\sum_{i=0}^{2^n-1} c_i |i\rangle = c_0 |00\ldots00\rangle + c_1 |00\ldots01\rangle + \ldots + c_{2^n-1} |11\ldots11\rangle , \tag{48}$$

with

$$\sum_{i=0}^{2^n-1} |c_i|^2 = 1.$$

If such a state can be represented as a tensor product of individual qubit states then the qubit states are *not entangled*. For example, it is easy to check that

$$\frac{1}{4\sqrt{2}} \left(\sqrt{3}\,|000\rangle + |001\rangle + 3\,|010\rangle + \sqrt{3}\,|011\rangle + \sqrt{3}\,|100\rangle + |101\rangle + 3\,|110\rangle + \sqrt{3}\,|111\rangle \right)$$

$$= \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \otimes \left(\frac{1}{2} |0\rangle + \frac{\sqrt{3}}{2} |1\rangle \right) \otimes \left(\frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle \right), \tag{49}$$

and, consequently, the quantum state is not entangled. On the other hand, an entangled state cannot be represented as a tensor product of individual qubit states.

For example, the 2-qubit state

$$\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle \tag{50}$$

does not allow a tensor product decomposition. Namely, for any $a, b, c, d \in \mathbb{C}$ such that $|a|^2 + |b|^2 = |c|^2 + |d|^2 = 1$, we have

$$\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle \neq (a\,|0\rangle + b\,|1\rangle) \otimes (c\,|0\rangle + d\,|1\rangle). \tag{51}$$

The two-qubit state $\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$ is known as one of the four maximally entangled Bell states. It can be constructed from the separable state $|00\rangle$

$$|00\rangle = (1 \cdot |0\rangle + 0 \cdot |1\rangle) \otimes (1 \cdot |0\rangle + 0 \cdot |1\rangle) \tag{52}$$

by applying the *Bell circuit* consisting of H and CNOT gates

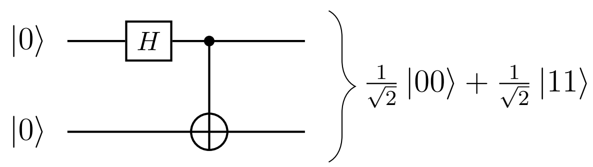

Figure 7: Bell circuit.

Running this circuit on the unentagled states $|01\rangle$, $|10\rangle$, and $|11\rangle$ will result in the construction of the other three Bell states:

$$|01\rangle \rightarrow \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \tag{53}$$

$$|10\rangle \rightarrow \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \tag{54}$$

$$|11\rangle \rightarrow \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \tag{55}$$

Entanglement can be achieved using other two-qubit gates as well, such as CZ and iSWAP.

Quantum mechanics postulates that for every measurable property of a physical system, there exists a corresponding self-adjoint operator, or, equivalently for a finite-dimensional case as the one of n qubits, a Hermitian operator. The expectione values of the physical observables correspond to the expectation values of the Hermitian operators. The expectation value $\langle L\rangle$ of the Hermitian operator L for a system in the state $|\psi\rangle$ is given by

$$\langle L\rangle = \langle\psi| L |\psi\rangle . \tag{56}$$

We would like to remind our readers that a linear operator L is Hermitian if $L^{\dagger} \equiv (L^{*})^{\top} = L$. All eigenvalues of a Hermitian operator are real. If an operator L is unitary, all of its eigenvalues lie on a unit circumference in the complex plane. If operator L is unitary and Hermitian, all its eigenvalues must be either $+1$ or -1.

Any 1-qubit Hermitian operator U can be represented as a linear combination of the identity operator I and Pauli operators X, Y and Z:

$$U = aI + bX + cY + dZ, \tag{57}$$

where coefficients a, b, c and d are real numbers. The expectation value of U applied to state $|\psi\rangle$ is given by the sum of expectation values of the individual terms

$$\langle\psi| U |\psi\rangle = a\langle\psi| I |\psi\rangle + b\langle\psi| X |\psi\rangle + c\langle\psi| Y |\psi\rangle + d\langle\psi| Z |\psi\rangle . \tag{58}$$

Therefore, in order to calculate the expectation value of a given 1-qubit Hermitian operator on a quantum computer, we can limit ourselves to the calculation of the expectation values of the operators I, X, Y and Z. Note that all of these operators are unitary Hermitian operators. Pauli operators X, Y and Z have eigenvalues $+1$ and -1. The identity operator I has only one eigenvalue $+1$ and any 1-qubit state is its eigenstate. We explain in what follows, a direct procedure to compute the desired expectation values. Note that the procedure illustrated represents a simplification of the "approximate quantum state" method (Paini *et al* 2021) actually employed in the experiment of Section 6. Providing a comprehensive explanation of the approximate quantum state technique would have been an unnecessary digression from the paper's objectives.

We start with the identity operator I. The case of the identity operator is trivial and no quantum circuit execution is required, since the expectation value of I is always 1:

$$\langle\psi| I |\psi\rangle = \langle\psi|\psi\rangle = 1. \tag{59}$$

We move on to the Z gate. In the computational basis, z-basis, $|\psi\rangle$ can be represented via a superposition of the basis states $|0\rangle$ and $|1\rangle$ as

$$|\psi\rangle = \alpha_z |0\rangle + \beta_z |1\rangle, \tag{60}$$

with $\alpha_z, \beta_z \in \mathbb{C}$. The expectation $\langle\psi| Z |\psi\rangle$ can therefore be expressed as

$$\begin{aligned}\langle\psi| Z |\psi\rangle &= |\alpha_z|^2 \langle 0| Z |0\rangle + \alpha_z^* \beta_z \langle 0| Z |1\rangle + \alpha_z \beta_z^* \langle 1| Z |0\rangle + |\beta_z|^2 \langle 1| Z |1\rangle \\ &= |\alpha_z|^2 \langle 0|0\rangle - \alpha_z^* \beta_z \langle 0|1\rangle + \alpha_z \beta_z^* \langle 1|0\rangle - |\beta_z|^2 \langle 1|1\rangle \\ &= |\alpha_z|^2 - |\beta_z|^2,\end{aligned} \tag{61}$$

where the definition of the Z gate and orthogonality of the basis states have been used for the second and third equality. The quantities $|\alpha_z|^2$ and $|\beta_z|^2$ are the probabilities that after the z-basis measurement, the quantum state $|\psi\rangle$ will be $|0\rangle$ or $|1\rangle$ respectively. In order to find these quantities, we can therefore run the quantum circuit to construct the state $|\psi\rangle$ and perform a measurement, repeating this N times and collecting statistics. The probability of finding a qubit in state $|0\rangle$ is then estimated as n_0/N, where n_0 is the number of state $|0\rangle$ measurements. Similarly, the probability of finding a qubit in state $|1\rangle$ can be estimated as n_1/N, where n_1 is the number of state $|1\rangle$ measurements. Therefore, the expectation value of Z in the state $|\psi\rangle$ is given by

$$\langle\psi| Z |\psi\rangle = \frac{n_0 - n_1}{N}. \tag{62}$$

The quantum state $|\psi\rangle$ can also be decomposed into a superposition of the basis states $\{|+i\rangle, |-i\rangle\}$ (y-basis) and $\{|+\rangle, |-\rangle\}$ (x-basis):

$$|\psi\rangle = \alpha_x |+\rangle + \beta_x |-\rangle = \alpha_y |+i\rangle + \beta_y |-i\rangle. \tag{63}$$

If we can perform measurement in the x-basis and the y-basis, the expectations $\langle\psi| X |\psi\rangle$ and $\langle\psi| Y |\psi\rangle$ can be calculated in exactly the same way as for the expectation $\langle\psi| Z |\psi\rangle$, namely

$$\langle\psi| X |\psi\rangle = \frac{n_+ - n_-}{N}, \quad \langle\psi| Y |\psi\rangle = \frac{n_{+i} - n_{-i}}{N}, \tag{64}$$

where n_+ and n_- are the numbers of measurements in the x-basis that correspond, respectively, to the $|+\rangle$ and $|-\rangle$ outcomes, and n_{+i} and n_{-i} are the numbers of measurements in the y-basis that correspond, respectively, to $|+i\rangle$ and $|-i\rangle$ outcomes. To measure in different bases, we do not need to change our measurement procedure and we can continue to measure in the z-basis. In fact, we can apply additional gates to $|\psi\rangle$ before measurement, such that the probability of measuring $|0\rangle$ in the z-basis after the execution of the additional gates is the same as the probability of measuring $|+\rangle$ in the x-basis in the state $|\psi\rangle$ (and, similarly, for $|+i\rangle$ in the y-basis).

Denoting the additional gates as H and G, respectively for the x-basis and y-basis, we have

$$H|\psi\rangle = H(\alpha_x|+\rangle + \beta_x|-\rangle) = \alpha_x|0\rangle + \beta_x|1\rangle, \tag{65}$$

with $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$ and

$$G|\psi\rangle = G(\alpha_y|+i\rangle + \beta_y|-i\rangle) = \alpha_y|0\rangle + \beta_y|1\rangle, \tag{66}$$

with $G|+i\rangle = |0\rangle$ and $G|-i\rangle = |1\rangle$. The operators H, known as the Hadamard gate, and G are given by

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \tag{67}$$

and

$$G = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -\mathrm{i} \\ 1 & \mathrm{i} \end{bmatrix} = HS^{\dagger}. \tag{68}$$

Interested readers can find detailed descriptions of a wider range of quantum gates, parameterised quantum circuits, and calculation of expectation values of the tensor products of Pauli operators on a quantum computer in (Jacquier and Kondratyev 2022).

B Quantum Feature Map

Figure 8 displays a quantum circuit that implements a quantum feature map: it maps each classical sample from the dataset into the corresponding quantum state. The circuit consists of four qubits shown in the figure as horizontal lines and several layers of 1-qubit and 2-qubit gates.

We encode data through 1-qubit gate operations, performed as rotations of the individual qubit states around the x, y and z axes. Each sample is a vector of four features, and each feature is mapped into a qubit's rotation angle. More specifically, the proposed angle encoding scheme is given by the following expression:

$$\theta_i(j) = \frac{\pi}{7}\left(\frac{F_i(j) - \min(F(j))}{\max(F(j)) - \min(F(j))}\right), \quad i = 1, \ldots, N_{samples}, \quad j = 1, \ldots, 4. \tag{69}$$

where $\min(F(j))$ and $\max(F(j))$ are the minimum and maximum values of feature $F(j)$ across all samples.

The 2-qubit gates are iSWAP gates. In total, we have six layers of iSWAP gates creating entanglement; alternating with seven layers of R_x, R_y and R_z gates performing repeated data encoding.

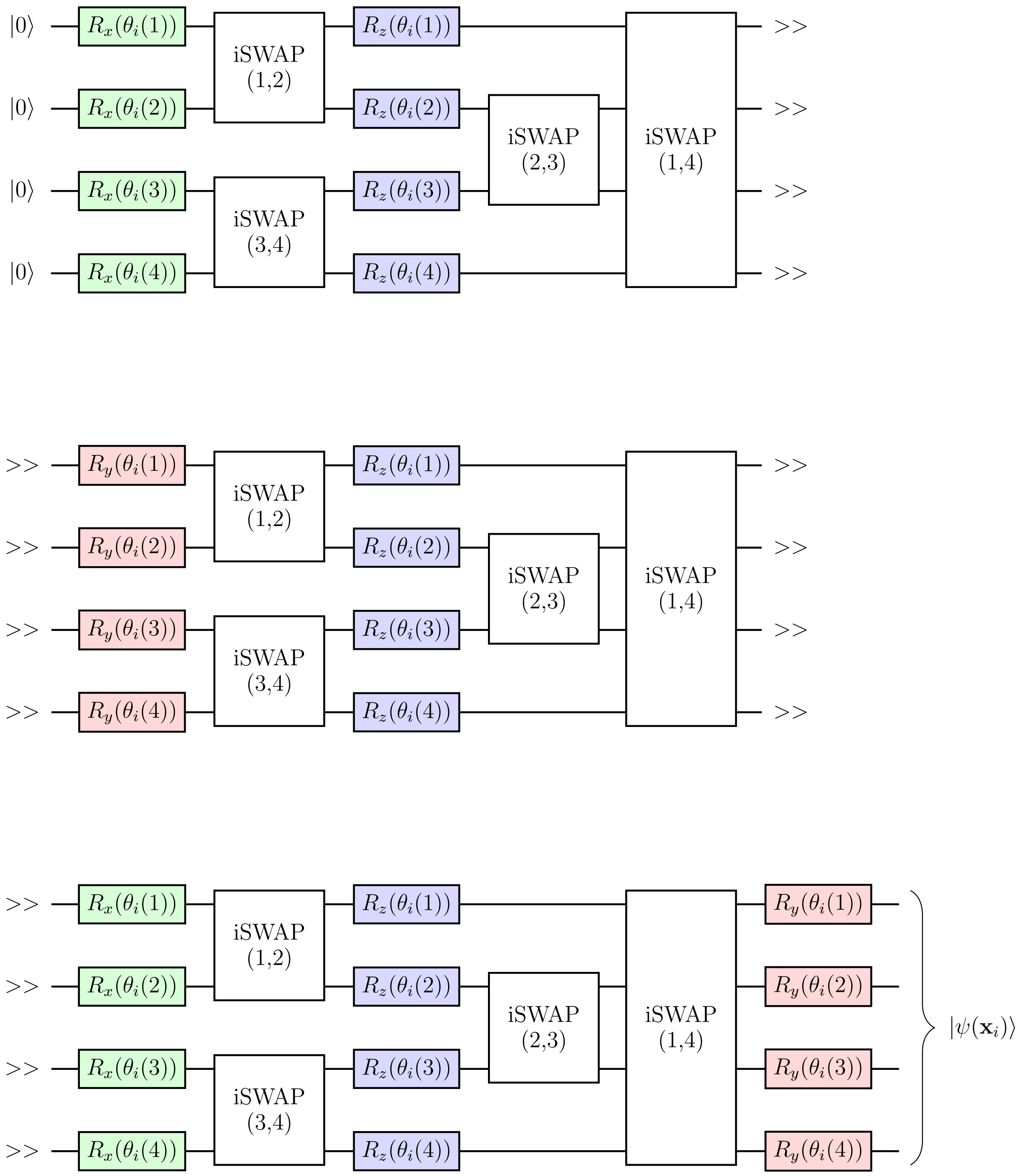

Figure 8: Quantum circuit (four qubits) implementing a quantum feature map.

C Estimation of Frobenius Distance on a Quantum Computer

Consider the space of linear operators $\mathcal{H}_\mathcal{O}$ with dimension $D_\mathcal{O} = D^2 = (2^n)^2$ acting in the Hilbert space $\mathcal{H}$ of a quantum system with dimension $D = 2^n$. A distance squared d^2 between two Hermitian operators p and q can be derived from a linear product $(\cdot, \cdot)$ defined in $\mathcal{H}_\mathcal{O}$:

$$d^2(p, q) = \|p - q\|^2 = (p - q, p - q). \tag{70}$$

If $\{B_i\}$ is a complete orthonormal set in $\mathcal{H}_{\mathcal{O}}$, then we can express p and q as vectors in $\mathbb{C}^{D_{\mathcal{O}}}$ with components $p_i := (B_i, p)$ and $q_i := (B_i, q)$ respectively. The Pauli operators $\{V_i := V_i^1 \otimes \ldots \otimes V_i^n\}$, with $V_i^j = \{I, X, Y, Z\}$ corresponding to the j-th single-qubit identity and Pauli operators, are an orthogonal set with the choice of the linear product $(L, M) = \mathrm{Tr}[L^\dagger M]$, which induces the Frobenius norm. The distance squared d^2 can be expressed as the Euclidean distance in $\mathbb{C}^{D_{\mathcal{O}}}$:

$$d^2(p,q) = \sum_{i=1}^{D_{\mathcal{O}}} |p_i - q_i|^2 = \|\mathbf{p} - \mathbf{q}\|_E^2, \tag{71}$$

with column vectors $\mathbf{p} := (p_1, \ldots, p_{D_{\mathcal{O}}})^\top$ and $\mathbf{q} := (q_1, \ldots, q_{D_{\mathcal{O}}})^\top$. Fixing the normalisation of the operators $\{V_i\}$, i.e., choosing $B_i = V_i/\sqrt{D}$ and p and q representing density matrices, the elements p_i and q_i become real numbers that can be expressed as

$$p_i = \frac{1}{\sqrt{D}} \mathrm{Tr}[V_i p] = \frac{1}{\sqrt{D}} \langle V_i \rangle_p \,, \quad q_i = \frac{1}{\sqrt{D}} \mathrm{Tr}[V_i q] = \frac{1}{\sqrt{D}} \langle V_i \rangle_q \,, \tag{72}$$

and we have

$$d^2(p,q) = \frac{1}{D} \sum_{i=1}^{D_{\mathcal{O}}} |\mathrm{Tr}[V_i p] - \mathrm{Tr}[V_i q]|^2 = \frac{1}{D} \sum_{i=1}^{D_{\mathcal{O}}} | \langle V_i \rangle_p - \langle V_i \rangle_q |^2. \tag{73}$$

The Frobenius distance as defined in Section 5 can be written as a Euclidean distance between vectors of expectation values of Paulis

$$d_F^2(p,q) = \frac{1}{4D} \sum_{i=1}^{D_{\mathcal{O}}} | \langle V_i \rangle_p - \langle V_i \rangle_q |^2 = \frac{1}{4D} \|\mathbf{P} - \mathbf{Q}\|_E^2, \tag{74}$$

where $\mathbf{P} := (\langle V_1 \rangle_p, \ldots, \langle V_{D_{\mathcal{O}}} \rangle_p)^\top$ and $\mathbf{Q} := (\langle V_1 \rangle_q, \ldots, \langle V_{D_{\mathcal{O}}} \rangle_q)^\top$.

The complete basis set of operators V_i consists of 4^n elements. This makes working with the complete basis set impractical or even impossible for large n. Therefore, we choose a subset of L operators V_i from the full set and approximate the sum on the expectation values of all Pauli operators with the sum on the subset of L Pauli operators:

$$\frac{1}{4D} \sum_{i=1}^{L} | \langle V_i \rangle_p - \langle V_i \rangle_q |^2 \leq \frac{1}{4D} \sum_{i=1}^{D_{\mathcal{O}}} | \langle V_i \rangle_p - \langle V_i \rangle_q |^2. \tag{75}$$

Thus, we can define the Frobenius distance estimator as

$$\tilde{d}_F(p,q) := \frac{1}{2} \sqrt{\frac{\sum_{i=1}^{L} | \langle V_i \rangle_p - \langle V_i \rangle_q |^2}{D}}. \tag{76}$$

If we consider the subspace of $\mathcal{H}_{\mathcal{O}}$ spanned by the L Pauli operators, then

$$\tilde{d}_F(p,q) := \frac{1}{2} \sqrt{\frac{\|\mathbf{P}_L - \mathbf{Q}_L\|_E^2}{D}}, \tag{77}$$

where $\mathbf{P}_L := (\langle V_1\rangle_p, \ldots, \langle V_L\rangle_p)^\top$ and $\mathbf{Q}_L := (\langle V_1\rangle_q, \ldots, \langle V_L\rangle_q)^\top$. We can interpret this as the exact Frobenius distance with a quantum feature map that does not map to the full density matrix, but to vectors of components of the full density matrix on a Pauli basis.

Algorithm 1 provides detailed instructions for obtaining the Frobenius distance estimate on a quantum computer.

Algorithm 1 Frobenius distance estimator for two datasets

Result: Frobenius distance.

1: Design n-qubit quantum circuit (quantum feature map) with layers of one-qubit gates (encoding sample) alternating with layers of two-qubit gates (creating entanglement). By default, the range of rotation angles should be set at $[0, \pi/m]$, where m is the number of data encoding layers.

2: Prepare two datasets (A and B) – calculate features, standardise features, etc.

3: For dataset A:

- For each sample $i = 1, \ldots, [A]$, run the quantum circuit N times measuring Pauli Z_k on all quantum registers $k = 1, \ldots, n$ ($+1$ if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). Calculate expectation value $\langle Z_k\rangle_i^A$ as the average value of measured Z_k across all quantum circuit runs.
- For each sample $i = 1, \ldots, [A]$, place H gate on **all** quantum registers before measurement and run the quantum circuit N times measuring Pauli X_k on all quantum registers $k = 1, \ldots, n$ ($+1$ if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). Calculate expectation value $\langle X_k\rangle_i^A$ as the average value of measured X_k across all quantum circuit runs.
- For each sample $i = 1, \ldots, [A]$, place $HS^\dagger$ gates on **all** quantum registers before measurement and run the quantum circuit N times measuring Pauli Y_k on all quantum registers $k = 1, \ldots, n$ ($+1$ if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). Calculate expectation value $\langle Y_k\rangle_i^A$ as the average value of measured Y_k across all quantum circuit runs.
- For each sample $i = 1, \ldots, [A]$, place either H or $HS^\dagger$ gates on **some** quantum registers before measurement and run the quantum circuit N times measuring either Pauli X or Pauli Y or Pauli Z on the corresponding quantum registers ($+1$ if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). If gate H was placed on quantum register k before measurement and gates $HS^\dagger$ were placed on quantum register l before measurement, we will measure X_k and Y_l; the expectation value $\langle X_k Y_l\rangle_i^A$ is then calculated as the average value of the product $X_k Y_l$ of measured values of X_k and Y_l across all quantum circuit runs.
- Note that in practice, as previously mentioned and as demonstrated for the experiments of Section 6, separate executions under different changes of bases can be avoided by utilising the approximate state method.

4: For dataset B:

- For each sample $j = 1, \ldots, [B]$, run the quantum circuit N times measuring Pauli Z_k on all quantum registers $k = 1, \ldots, n$ ($+1$ if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). Calculate expectation value $\langle Z_k \rangle_j^B$ as the average value of measured Z_k across all quantum circuit runs.
- For each sample $j = 1, \ldots, [B]$, place H gate on **all** quantum registers before measurement and run the quantum circuit N times measuring Pauli X_k on all quantum registers $k = 1, \ldots, n$ ($+1$ if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). Calculate expectation value $\langle X_k \rangle_j^B$ as the average value of measured X_k across all quantum circuit runs.
- For each sample $j = 1, \ldots, [B]$, place $HS^\dagger$ gates on **all** quantum registers before measurement and run the quantum circuit N times measuring Pauli Y_k on all quantum registers $k = 1, \ldots, n$ ($+1$ if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). Calculate expectation value $\langle Y_k \rangle_j^B$ as the average value of measured Y_k across all quantum circuit runs.
- For each sample $j = 1, \ldots, [B]$, place either H or $HS^\dagger$ gates on **some** quantum registers before measurement and run the quantum circuit N times measuring either Pauli X or Pauli Y or Pauli Z on the corresponding quantum registers ($+1$ if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). If gate H was placed on quantum register k before measurement and gates $HS^\dagger$ were placed on quantum register l before measurement, we will measure X_k and Y_l; the expectation value $\langle X_k Y_l \rangle_j^B$ is then calculated as the average value of the product $X_k Y_l$ of measured values of X_k and Y_l across all quantum circuit runs.

5: Without loss of generality, let us assume that we measured $3n$ second order Paulis (i.e., the same number as all first order Paulis). We have now two arrays: $[A] \times 6n$ array of expectation values of first and second order Paulis for dataset A and $[B] \times 6n$ array of expectation values of first and second order Paulis for dataset B. For every column of these arrays we calculate the average expectation values across all samples in the datasets:

$$\overline{\langle X_1 \rangle^A} = \frac{1}{[A]} \sum_{i=1}^{[A]} \langle X_1 \rangle_i^A, \ldots; \quad \overline{\langle X_1 \rangle^B} = \frac{1}{[B]} \sum_{j=1}^{[B]} \langle X_1 \rangle_j^B, \ldots$$

6: We have now two $6n$-dimensional column vectors for datasets A and B:

$$\left(\overline{\langle X_1 \rangle^A}, \overline{\langle X_2 \rangle^A}, \ldots\right)^\top; \quad \left(\overline{\langle X_1 \rangle^B}, \overline{\langle X_2 \rangle^B}, \ldots\right)^\top.$$

The next step is to calculate the column vector of differences:

$$\mathbf{c} = \left(\overline{\langle X_1 \rangle^A} - \overline{\langle X_1 \rangle^B}, \overline{\langle X_2 \rangle^A} - \overline{\langle X_2 \rangle^B}, \ldots\right)^\top.$$

7: Calculate the estimate of the Frobenius distance:

$$\tilde{d}_F(A, B) = \frac{1}{2} \sqrt{\frac{\mathbf{c}^\top \mathbf{c}}{2^n}}.$$

D Comparison of Cumulative Distribution Functions

Figure 3 displays the mean distances, along with their standard deviations, for the classical and quantum MMDs calculated from 100 simulated scenarios of assets A, B and C.

Figure 9 provides additional information on the individual distances obtained. The top plot presents distances calculated using an MMD with classical Gaussian kernel. The bottom plot contains results from an MMD using a quantum feature map with four qubits as described in Appendix B and executed on the Rigetti Ankaa-2 quantum processor. Each plot consists of a cumulative distribution function $CDF(X)$ specifying the percentage of distances which are less than or equal to the parameter X. In addition, the distances themselves are plotted horizontally as another visual indicator of the spread of the distribution.

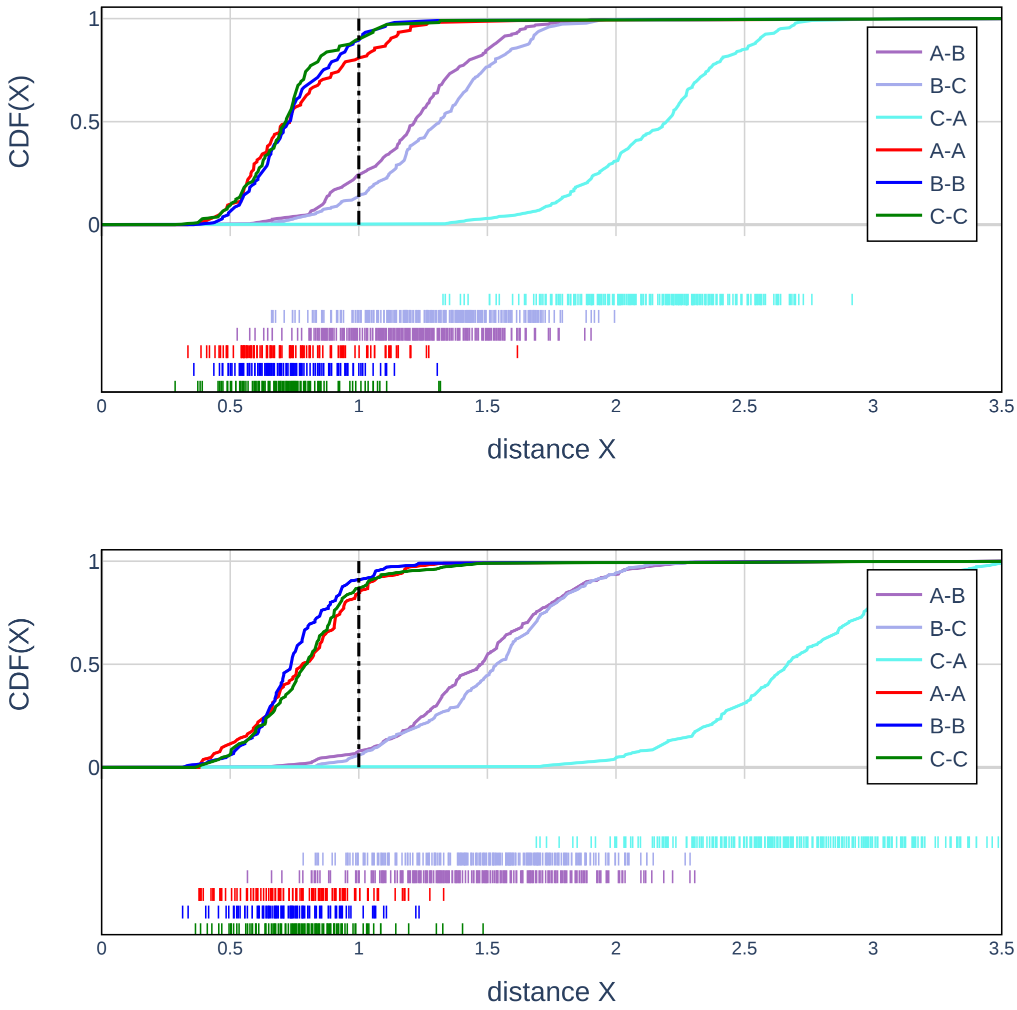

Figure 9: Comparison of classical (top) and quantum (bottom) distances for 100 simulated datasets A ($\varrho = 0$), B ($\varrho = 0.2$) and C ($\varrho = 0.4$). The quantum results are obtained from the Rigetti Ankaa-2 quantum processor.

Distances between realisations of two different asset datasets are displayed for A-B (purple), B-C (grey-blue) and C-A (turquoise). Distances between two realisations of the same asset dataset are displayed for A-A (red), B-B (blue) and C-C (green). The dashed black line represents the distance level threshold. Distances below the threshold indicate that the two datasets are likely to come from the same distribution, while distances above the threshold indicate that the two datasets are likely to come from different distributions. It can be seen that distances between realisations of two different datasets are shifted further to the right for the quantum MMD (bottom plot). This results in it providing improved classification accuracy.